

Linearising Explicit Substitutions using Intersection Types

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Motivation

- ▶ The λ -calculus allows for unlimited duplication and erasing of variables
 $\Rightarrow$ lacks control over resource usage
- ▶ There are resource-aware models
 - ▶ linear calculi
 - ▶ resource-aware calculi
 - ▶ λ -calculus with multiplicities
 - ▶ quantitative types
- ▶ How can we relate the λ -calculus with resource-aware models?

Intersection Types (briefly)

Definition

Let α range over an infinite set of type variables

$$(\cap\text{-types}) \quad \sigma ::= \alpha \mid \sigma_1 \cap \cdots \cap \sigma_n \rightarrow \sigma$$

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Example

$$\lambda x.xx : (\alpha \rightarrow \beta) \cap \alpha \rightarrow \beta$$

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Example

$$\lambda x.xx : (\alpha \rightarrow \beta) \cap \alpha \rightarrow \beta$$

ACI-intersection types

$$\sigma \cap \sigma = \sigma$$

AC-intersection types

$$\sigma \cap \sigma \neq \sigma$$

Linearisation through Term Expansion

- Relates λ -terms with linear terms

Mário Florido and Luís Damas. “Linearization of the lambda-calculus and its relation with intersection type systems”. In: *Journal of Functional Programming* 14.5 (2004)

Linearisation through Term Expansion

Given the term $(\lambda x.xx)I$ and the type $\sigma \equiv \alpha \rightarrow \alpha$.

$$\begin{array}{c}
 T \quad \frac{\mathcal{E}(x : \sigma \rightarrow \sigma) \triangleleft (x_1, \{x : \{x_1 : \sigma \rightarrow \sigma\}\}) \quad \mathcal{E}(x : \sigma) \triangleleft (x_2, \{x : \{x_2 : \sigma\}\})}{\mathcal{E}(xx : \sigma) \triangleleft (x_1 x_2, \{x : \{x_1 : \sigma \rightarrow \sigma, x_2 : \sigma\}\})} \\
 \frac{\mathcal{E}(xx : \sigma) \triangleleft (x_1 x_2, \{x : \{x_1 : \sigma \rightarrow \sigma, x_2 : \sigma\}\})}{\mathcal{E}(\lambda x.xx : (\sigma \rightarrow \sigma) \cap \sigma \rightarrow \sigma) \triangleleft \lambda x_1 x_2. x_1 x_2} \\
 \\
 T \quad \frac{\begin{array}{c} \dots \\ \mathcal{E}(I : \sigma \rightarrow \sigma) \triangleleft I \end{array} \quad \begin{array}{c} \dots \\ \mathcal{E}(I : \sigma) \triangleleft I \end{array}}{\mathcal{E}((\lambda x.xx)I : \sigma) \triangleleft (\lambda x_1 x_2. x_1 x_2)II}
 \end{array}$$

Mário Florido and Luís Damas. “Linearization of the lambda-calculus and its relation with intersection type systems”. In: *Journal of Functional Programming* 14.5 (2004)

Term Expansion

- ▶ *Term expansion* is guided by intersection types
- ▶ Different notions of intersection $\Rightarrow$ different target languages

Explicit Substitutions

- Explicit substitutions are part of the term language

Kristoffer Rose. “Explicit substitution: tutorial & survey”. In: *Explicit Substitution – Tutorial & Survey BRICS Basic Research in Computer Science* (1996)

Explicit Substitutions

- Explicit substitutions are part of the term language

Example

$$\begin{array}{lcl} (\lambda x.xx)I & \xrightarrow[b]{} & (xx)\langle x := I \rangle \\ & \xrightarrow[xap]{} & x\langle x := I \rangle x\langle x := I \rangle \\ & \xrightarrow[xv]{} & Ix\langle x := I \rangle \\ & \xrightarrow[xv]{} & II \\ & \xrightarrow[b]{} & z\langle z := I \rangle \\ & \xrightarrow[xv]{} & I \end{array}$$

Kristoffer Rose. “Explicit substitution: tutorial & survey”. In: *Explicit Substitution – Tutorial & Survey BRICS Basic Research in Computer Science* (1996)

Explicit Substitutions

- Explicit substitutions are part of the term language

Example $(\lambda x.xx)I \xrightarrow[b]{\quad} (xx)\langle x := I \rangle$
 $\xrightarrow[xap]{\quad} x\langle x := I \rangle x\langle x := I \rangle$
 $\xrightarrow[xv]{\quad} Ix\langle x := I \rangle$
 $\xrightarrow[xv]{\quad} II$
 $\xrightarrow[b]{\quad} z\langle z := I \rangle$
 $\xrightarrow[xv]{\quad} I$

calculus was
modified



weak-head
reductions

Kristoffer Rose. “Explicit substitution: tutorial & survey”. In: *Explicit Substitution – Tutorial & Survey BRICS Basic Research in Computer Science* (1996)

Goal

- ▶ Apply *term expansion* to a calculus with explicit substitutions

For the standard λ -calculus

ACI-intersection types

(associative, commutative and idempotent)



terms typed in the simple type system

AC-intersection types

(associative, commutative and non-idempotent)



affine and linear terms

Sandra Alves and Mário Florido. “Structural Rules and Algebraic Properties of Intersection Types”. In: *International Colloquium on Theoretical Aspects of Computing*. Springer. 2022

For the λ -calculus with explicit substitutions

ACI-intersection types

(associative, commutative and idempotent)



?

AC-intersection types

(associative, commutative and non-idempotent)



?

For the λ -calculus with explicit substitutions

ACI-intersection types

(associative, commutative and idempotent)



λ -calculus with infinite *multiplicities*

AC-intersection types

(associative, commutative and non-idempotent)



λ -calculus with finite *multiplicities*

Multiplicities

- Gérard Boudol proposes the *λ -calculus with multiplicities*

Definition (Boudol's terms)

M	$::=$	$x \mid \lambda x.M \mid (MP) \mid M\langle P/x \rangle$	terms
P	$::=$	$1 \mid M \mid (P \mid P) \mid M^{\textcolor{red}{m}}$	bags of terms
V	$::=$	$\lambda x.M \mid V\langle P/x \rangle$	values

- multiplicity $\textcolor{red}{m} \in \mathbb{N} \cup \{\infty\}$

Example

$$((\lambda x.(xx^1))/^2)$$

Example

$$((\lambda x.(xx^1))I^2) \rightarrow_B (xx^1)\langle I^2/x \rangle$$

Example

$$\begin{aligned} ((\lambda x. (xx^1)) l^2) &\rightarrow_B (xx^1) \langle l^2 / x \rangle \\ &\rightarrow_B (l x^1) \langle l^1 / x \rangle \end{aligned}$$

Example

$$\begin{aligned} ((\lambda x.(xx^1))l^2) &\rightarrow_B (xx^1)\langle l^2/x \rangle \\ &\rightarrow_B (lx^1)\langle l^1/x \rangle \\ &\rightarrow_B (z\langle x^1/z \rangle)\langle l^1/x \rangle \end{aligned}$$

Example

$$\begin{aligned} ((\lambda x. (xx^1))l^2) &\rightarrow_B (xx^1)\langle l^2/x \rangle \\ &\rightarrow_B (lx^1)\langle l^1/x \rangle \\ &\rightarrow_B (z\langle x^1/z \rangle)\langle l^1/x \rangle \\ &\rightarrow_B (x\langle 1/z \rangle)\langle l^1/x \rangle \end{aligned}$$

Example

$$\begin{aligned}((\lambda x.(xx^1))l^2) &\rightarrow_B (xx^1)\langle l^2/x \rangle \\ &\rightarrow_B (lx^1)\langle l^1/x \rangle \\ &\rightarrow_B (z\langle x^1/z \rangle)\langle l^1/x \rangle \\ &\rightarrow_B (x\langle 1/z \rangle)\langle l^1/x \rangle \\ &\rightarrow_B (l\langle 1/x \rangle)\langle 1/z \rangle\end{aligned}$$

Example

$$\begin{aligned}((\lambda x.(xx^1))I^2) &\rightarrow_B (xx^1)\langle I^2/x \rangle \\ &\rightarrow_B (Ix^1)\langle I^1/x \rangle \\ &\rightarrow_B (z\langle x^1/z \rangle)\langle I^1/x \rangle \\ &\rightarrow_B (x\langle 1/z \rangle)\langle I^1/x \rangle \\ &\rightarrow_B (I\langle 1/x \rangle)\langle 1/z \rangle \\ &\equiv I\end{aligned}$$

Example

$$\begin{aligned}
 ((\lambda x. (xx^1))I^2) &\rightarrow_B (xx^1)\langle I^2/x \rangle \\
 &\rightarrow_B (Ix^1)\langle I^1/x \rangle \\
 &\rightarrow_B (z\langle x^1/z \rangle)\langle I^1/x \rangle \\
 &\rightarrow_B (x\langle 1/z \rangle)\langle I^1/x \rangle \\
 &\rightarrow_B (I\langle 1/x \rangle)\langle 1/z \rangle \\
 &\equiv I
 \end{aligned}$$

$$\begin{aligned}
 ((\lambda x. (xx^\infty))I^\infty) &\rightarrow_B (xx^\infty)\langle I^\infty/x \rangle \\
 &\rightarrow_B (Ix^\infty)\langle I^\infty/x \rangle \\
 &\rightarrow_B (z\langle x^\infty/z \rangle)\langle I^\infty/x \rangle \\
 &\rightarrow_B (x\langle x^\infty/z \rangle)\langle I^\infty/x \rangle \\
 &\rightarrow_B (I\langle I^\infty/x \rangle)\langle x^\infty/z \rangle \\
 &\rightarrow_B I
 \end{aligned}$$

Term Expansion with ACI-intersection types

Consider the the term $(\lambda x.xx)I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\overline{\mathcal{E}^\infty((\lambda x.xx)I : \sigma) \triangleleft (M_0, A_0)}$$

Term Expansion with ACI-intersection types

Consider the the term $(\lambda x.xx)I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\frac{\mathcal{E}^\infty(\lambda x.xx : (\sigma \rightarrow \sigma) \cap \sigma \rightarrow \sigma) \triangleleft (M_1, A_1)}{\mathcal{E}^\infty((\lambda x.xx)I : \sigma) \triangleleft (M_0, A_0)} \quad T_1 \quad T_2$$

Term Expansion with ACI-intersection types

Consider the the term $(\lambda x.xx)I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\frac{\frac{\mathcal{E}^\infty(xx : \sigma) \triangleleft (M_2, A_2 \ \& \ \{x : (\sigma \rightarrow \sigma) \cap \sigma\})}{\mathcal{E}^\infty(\lambda x.xx : (\sigma \rightarrow \sigma) \cap \sigma \rightarrow \sigma) \triangleleft (M_1, A_1)} \quad T_1 \quad T_2}{\mathcal{E}^\infty((\lambda x.xx)I : \sigma) \triangleleft (M_0, A_0)}$$

Term Expansion with ACI-intersection types

Consider the the term $(\lambda x.xx)I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\begin{array}{c}
 \frac{\mathcal{E}^\infty(x : \sigma \rightarrow \sigma) \triangleleft (x, \{x : \sigma \rightarrow \sigma\}) \quad \mathcal{E}^\infty(x : \sigma) \triangleleft (x, \{x : \sigma\})}{\mathcal{E}^\infty(xx : \sigma) \triangleleft (M_2, A_2 \ \& \ \{x : (\sigma \rightarrow \sigma) \cap \sigma\})} \\
 \frac{\mathcal{E}^\infty(\lambda x.xx : (\sigma \rightarrow \sigma) \cap \sigma \rightarrow \sigma) \triangleleft (M_1, A_1)}{\mathcal{E}^\infty((\lambda x.xx)I : \sigma) \triangleleft (M_0, A_0)} \quad T_1 \quad T_2
 \end{array}$$

Term Expansion with ACI-intersection types

Consider the the term $(\lambda x.xx)I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\begin{array}{c}
 \frac{\mathcal{E}^\infty(x : \sigma \rightarrow \sigma) \triangleleft (x, \{x : \sigma \rightarrow \sigma\}) \quad \mathcal{E}^\infty(x : \sigma) \triangleleft (x, \{x : \sigma\})}{\mathcal{E}^\infty(xx : \sigma) \triangleleft ((\textcolor{red}{xx}^\infty), \{x : (\sigma \rightarrow \sigma) \cap \sigma\})} \\
 \frac{\mathcal{E}^\infty(\lambda x.xx : (\sigma \rightarrow \sigma) \cap \sigma \rightarrow \sigma) \triangleleft (M_1, A_1)}{\mathcal{E}^\infty((\lambda x.xx)I : \sigma) \triangleleft (M_0, A_0)} \quad T_1 \quad T_2
 \end{array}$$

Term Expansion with ACI-intersection types

Consider the the term $(\lambda x.xx)I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

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 \frac{\mathcal{E}^\infty(\lambda x.xx : (\sigma \rightarrow \sigma) \cap \sigma \rightarrow \sigma) \triangleleft \lambda x.(xx)^\infty}{\mathcal{E}^\infty((\lambda x.xx)I : \sigma) \triangleleft (M_0, A_0)} \quad T_1 \quad T_2
 \end{array}$$

Term Expansion with ACI-intersection types

Consider the the term $(\lambda x.xx)I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$T_1 \quad \frac{}{\mathcal{E}^\infty(I : \sigma \rightarrow \sigma) \triangleleft (M_3, A_3)}$$

$$T_2 \quad \frac{}{\mathcal{E}^\infty(I : \sigma) \triangleleft (M_4, A_4)}$$

$$\frac{\frac{\mathcal{E}^\infty(x : \sigma \rightarrow \sigma) \triangleleft (x, \{x : \sigma \rightarrow \sigma\}) \quad \mathcal{E}^\infty(x : \sigma) \triangleleft (x, \{x : \sigma\})}{\mathcal{E}^\infty(xx : \sigma) \triangleleft ((xx^\infty), \{x : (\sigma \rightarrow \sigma) \cap \sigma\})}}{\frac{\mathcal{E}^\infty(\lambda x.xx : (\sigma \rightarrow \sigma) \cap \sigma \rightarrow \sigma) \triangleleft \lambda x.(xx^\infty)}{\mathcal{E}^\infty((\lambda x.xx)I : \sigma) \triangleleft (M_0, A_0)}} \quad T_1 \quad T_2$$

Term Expansion with ACI-intersection types

Consider the the term $(\lambda x.xx)I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\boxed{T_1} \quad \frac{\mathcal{E}^\infty(z : \sigma) \triangleleft (z, \{z : \sigma\})}{\mathcal{E}^\infty(I : \sigma \rightarrow \sigma) \triangleleft (M_3, A_3)}$$

$$\boxed{T_2} \quad \frac{\mathcal{E}^\infty(z : \alpha) \triangleleft (z, \{z : \alpha\})}{\mathcal{E}^\infty(I : \sigma) \triangleleft (M_4, A_4)}$$

$$\frac{\frac{\mathcal{E}^\infty(x : \sigma \rightarrow \sigma) \triangleleft (x, \{x : \sigma \rightarrow \sigma\}) \quad \mathcal{E}^\infty(x : \sigma) \triangleleft (x, \{x : \sigma\})}{\mathcal{E}^\infty(xx : \sigma) \triangleleft ((xx)^\infty, \{x : (\sigma \rightarrow \sigma) \cap \sigma\})}}{\frac{\mathcal{E}^\infty(\lambda x.xx : (\sigma \rightarrow \sigma) \cap \sigma \rightarrow \sigma) \triangleleft \lambda x.(xx)^\infty}{\mathcal{E}^\infty((\lambda x.xx)I : \sigma) \triangleleft (M_0, A_0)}} \quad T_1 \quad T_2$$

Term Expansion with ACI-intersection types

Consider the the term $(\lambda x.xx)I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\boxed{T_1} \quad \frac{\mathcal{E}^\infty(z : \sigma) \triangleleft (z, \{z : \sigma\})}{\mathcal{E}^\infty(I : \sigma \rightarrow \sigma) \triangleleft \textcolor{red}{I}}$$

$$\boxed{T_2} \quad \frac{\mathcal{E}^\infty(z : \alpha) \triangleleft (z, \{z : \alpha\})}{\mathcal{E}^\infty(I : \sigma) \triangleleft \textcolor{red}{I}}$$

$$\frac{\frac{\mathcal{E}^\infty(x : \sigma \rightarrow \sigma) \triangleleft (x, \{x : \sigma \rightarrow \sigma\}) \quad \mathcal{E}^\infty(x : \sigma) \triangleleft (x, \{x : \sigma\})}{\mathcal{E}^\infty(xx : \sigma) \triangleleft ((xx)^\infty, \{x : (\sigma \rightarrow \sigma) \cap \sigma\})}}{\frac{\mathcal{E}^\infty(\lambda x.xx : (\sigma \rightarrow \sigma) \cap \sigma \rightarrow \sigma) \triangleleft \lambda x.(xx)^\infty}{\mathcal{E}^\infty((\lambda x.xx)I : \sigma) \triangleleft (M_0, A_0)}} \quad T_1 \quad T_2$$

Term Expansion with ACI-intersection types

Consider the the term $(\lambda x.xx)I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\boxed{T_1} \quad \frac{\mathcal{E}^\infty(z : \sigma) \triangleleft (z, \{z : \sigma\})}{\mathcal{E}^\infty(I : \sigma \rightarrow \sigma) \triangleleft I}$$

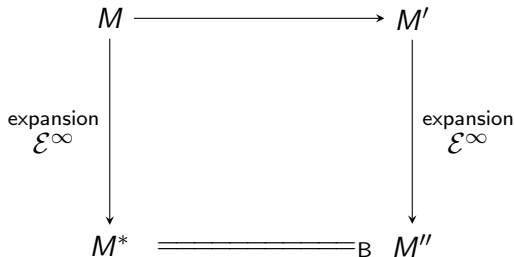
$$\boxed{T_2} \quad \frac{\mathcal{E}^\infty(z : \alpha) \triangleleft (z, \{z : \alpha\})}{\mathcal{E}^\infty(I : \sigma) \triangleleft I}$$

$$\frac{\frac{\mathcal{E}^\infty(x : \sigma \rightarrow \sigma) \triangleleft (x, \{x : \sigma \rightarrow \sigma\}) \quad \mathcal{E}^\infty(x : \sigma) \triangleleft (x, \{x : \sigma\})}{\mathcal{E}^\infty(xx : \sigma) \triangleleft ((xx)^\infty, \{x : (\sigma \rightarrow \sigma) \cap \sigma\})}}{\frac{\mathcal{E}^\infty(\lambda x.xx : (\sigma \rightarrow \sigma) \cap \sigma \rightarrow \sigma) \triangleleft \lambda x.(xx)^\infty}{\mathcal{E}^\infty((\lambda x.xx)I : \sigma) \triangleleft ((\lambda x.(xx)^\infty)I)^\infty}} \quad T_1 \quad T_2$$

Expansion and Infinite Multiplicities

Theorem

Given a λx -term M and an ACI-intersection type σ , such that $\mathcal{E}^\infty(M : \sigma) \triangleleft (M^*, A_1)$, if $M \xrightarrow[bxgc]{\quad} M'$ then $M^* =_B M''$ and $\mathcal{E}^\infty(M' : \sigma) \triangleleft (M'', A_2)$, where $A_2 \subseteq A_1$.



Illustrative Example

$$((\lambda x.xx)I)$$

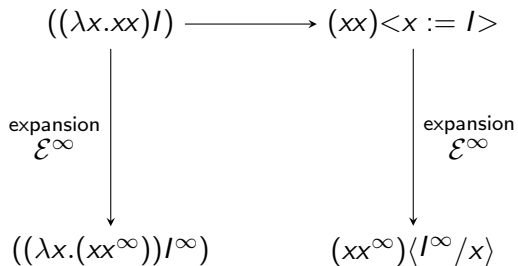
Illustrative Example

$$((\lambda x.xx)I) \longrightarrow (xx)\langle x := I \rangle$$

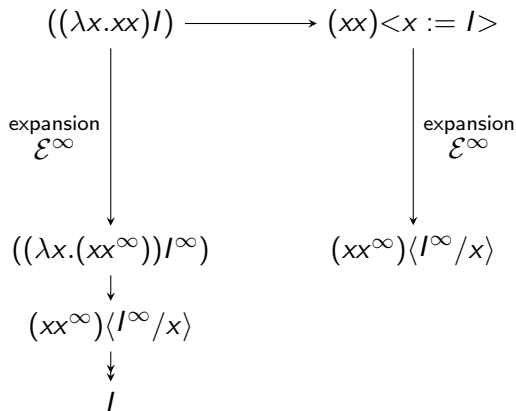
Illustrative Example

$$\begin{array}{ccc} ((\lambda x. xx)I) & \longrightarrow & (xx) \langle x := I \rangle \\ \downarrow \text{expansion } \mathcal{E}^\infty & & \\ ((\lambda x. (xx^\infty))I^\infty) & & \end{array}$$

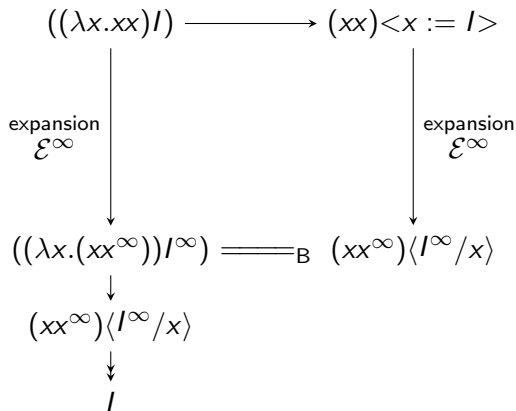
Illustrative Example



Illustrative Example



Illustrative Example



Term Expansion with AC-intersection types

Consider the the term $(\lambda f x. f(fx))I$, where $I \equiv \lambda z. z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\overline{\mathcal{E}((\lambda f x. f(fx))I : \sigma) \triangleleft (M_0, A_0)}$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\frac{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1)}{\mathcal{E}((\lambda fx.f(fx))I : \sigma) \triangleleft (M_0, A_0)} T_1$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\begin{array}{c}
 \mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft (M_2, A_2) \quad T_2 \quad T_3 \\
 \hline
 \mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1) \quad T_1 \\
 \hline
 \mathcal{E}((\lambda fx.f(fx))II : \sigma) \triangleleft (M_0, A_0)
 \end{array}$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\begin{array}{c}
 \mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (M_3, A_3 \otimes \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\}) \\
 \hline
 \mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft (M_2, A_2) \quad T_2 \quad T_3 \\
 \hline
 \mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1) \quad T_1 \\
 \hline
 \mathcal{E}((\lambda fx.f(fx))I : \sigma) \triangleleft (M_0, A_0)
 \end{array}$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\begin{array}{c}
 \mathcal{E}(f(fx) : \sigma) \triangleleft (M_4, A_4 \otimes \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\}) \\
 \hline
 \mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (M_3, A_3 \otimes \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\}) \\
 \hline
 \mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft (M_2, A_2) \quad T_2 \quad T_3 \\
 \hline
 \mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1) \quad T_1 \\
 \hline
 \mathcal{E}((\lambda fx.f(fx))II : \sigma) \triangleleft (M_0, A_0)
 \end{array}$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\begin{array}{c}
 \mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\}) \quad T_4 \\
 \hline
 \mathcal{E}(f(fx) : \sigma) \triangleleft (M_4, A_4 \otimes \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\}) \\
 \hline
 \mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (M_3, A_3 \otimes \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\}) \\
 \hline
 \mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft (M_2, A_2) \quad T_2 \quad T_3 \\
 \hline
 \mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1) \quad T_1 \\
 \hline
 \mathcal{E}((\lambda fx.f(fx))II : \sigma) \triangleleft (M_0, A_0)
 \end{array}$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

T_4

$$\frac{}{\mathcal{E}(fx : \sigma) \triangleleft (M_5, A_5 \otimes \{f : \sigma \rightarrow \sigma, x : \sigma\})}$$

$$\frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\})}{\mathcal{E}(f(fx) : \sigma) \triangleleft (M_4, A_4 \otimes \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})} \quad T_4$$

$$\frac{\mathcal{E}(f(fx) : \sigma) \triangleleft (M_4, A_4 \otimes \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})}{\mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (M_3, A_3 \otimes \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\})}$$

$$\frac{\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft (M_2, A_2)}{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1)} \quad T_2 \quad T_3$$

$$\frac{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1)}{\mathcal{E}((\lambda fx.f(fx))II : \sigma) \triangleleft (M_0, A_0)} \quad T_1$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\begin{array}{c}
 \boxed{T_4} \quad \frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\}) \quad \mathcal{E}(x : \sigma) \triangleleft (x, \{x : \sigma\})}{\mathcal{E}(fx : \sigma) \triangleleft (M_5, A_5 \otimes \{f : \sigma \rightarrow \sigma, x : \sigma\})} \\
 \\
 \frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\}) \quad T_4}{\mathcal{E}(f(fx) : \sigma) \triangleleft (M_4, A_4 \otimes \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})} \\
 \frac{\mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (M_3, A_3 \otimes \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\})}{\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft (M_2, A_2)} \quad T_2 \quad T_3 \\
 \frac{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1)}{\mathcal{E}((\lambda fx.f(fx))I : \sigma) \triangleleft (M_0, A_0)} \quad T_1
 \end{array}$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\begin{array}{c}
 \boxed{T_4} \quad \frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\}) \quad \mathcal{E}(x : \sigma) \triangleleft (x, \{x : \sigma\})}{\mathcal{E}(fx : \sigma) \triangleleft ((fx^1), \{f : \sigma \rightarrow \sigma, x : \sigma\})} \\
 \\
 \frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\}) \quad T_4}{\mathcal{E}(f(fx) : \sigma) \triangleleft (M_4, A_4 \otimes \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})} \\
 \frac{\mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (M_3, A_3 \otimes \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\})}{\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft (M_2, A_2)} \quad T_2 \quad T_3 \\
 \frac{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1)}{\mathcal{E}((\lambda fx.f(fx))II : \sigma) \triangleleft (M_0, A_0)} \quad T_1
 \end{array}$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\begin{array}{c}
 \boxed{T_4} \quad \frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\}) \quad \mathcal{E}(x : \sigma) \triangleleft (x, \{x : \sigma\})}{\mathcal{E}(fx : \sigma) \triangleleft ((fx^1), \{f : \sigma \rightarrow \sigma, x : \sigma\})} \\
 \\
 \frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\}) \quad T_4}{\mathcal{E}(f(fx) : \sigma) \triangleleft ((f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})} \\
 \frac{\mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (M_3, A_3 \otimes \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\})}{\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft (M_2, A_2)} \quad T_2 \quad T_3 \\
 \frac{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1)}{\mathcal{E}((\lambda fx.f(fx))II : \sigma) \triangleleft (M_0, A_0)} \quad T_1
 \end{array}$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\begin{array}{c}
 \frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\}) \quad T_4}{\mathcal{E}(f(fx) : \sigma) \triangleleft ((f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})} \\
 \frac{\mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (\lambda x.(f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\})}{\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft (M_2, A_2) \quad T_2 \quad T_3} \\
 \frac{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1) \quad T_1}{\mathcal{E}((\lambda fx.f(fx))I : \sigma) \triangleleft (M_0, A_0)}
 \end{array}$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\begin{array}{c}
 \frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\}) \quad T_4}{\mathcal{E}(f(fx) : \sigma) \triangleleft ((f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})} \\
 \frac{\mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (\lambda x.(f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\})}{\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft \lambda fx.(f(fx^1)^1) \quad T_2 \quad T_3} \\
 \frac{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1)}{\mathcal{E}((\lambda fx.f(fx))I : \sigma) \triangleleft (M_0, A_0) \quad T_1}
 \end{array}$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

T_2

$$\frac{}{\mathcal{E}(I : \sigma \rightarrow \sigma) \triangleleft (M_6, A_6)}$$

T_3

$$\frac{}{\mathcal{E}(I : \sigma \rightarrow \sigma) \triangleleft (M_7, A_7)}$$

$$\frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\})}{\mathcal{E}(f(fx) : \sigma) \triangleleft ((f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})} \quad T_4$$

$$\frac{\mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (\lambda x.(f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\})}{\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft \lambda fx.(f(fx^1)^1)} \quad T_2 \quad T_3$$

$$\frac{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1)}{\mathcal{E}((\lambda fx.f(fx))I : \sigma) \triangleleft (M_0, A_0)} \quad T_1$$

$$\mathcal{E}((\lambda fx.f(fx))I : \sigma) \triangleleft (M_0, A_0)$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

T_2

$$\frac{\mathcal{E}(z : \sigma) \triangleleft (z, \{z : \sigma\})}{\mathcal{E}(I : \sigma \rightarrow \sigma) \triangleleft (M_6, A_6)}$$

T_3

$$\frac{\mathcal{E}(z : \sigma) \triangleleft (z, \{z : \sigma\})}{\mathcal{E}(I : \sigma \rightarrow \sigma) \triangleleft (M_7, A_7)}$$

$$\frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\})}{\mathcal{E}(f(fx) : \sigma) \triangleleft ((f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})} \quad T_4$$

$$\frac{\mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (\lambda x.(f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\})}{\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft \lambda fx.(f(fx^1)^1)} \quad T_2 \quad T_3$$

$$\frac{\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft \lambda fx.(f(fx^1)^1)}{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1)} \quad T_1$$

$$\frac{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1)}{\mathcal{E}((\lambda fx.f(fx))I : \sigma) \triangleleft (M_0, A_0)}$$

$$\mathcal{E}((\lambda fx.f(fx))I : \sigma) \triangleleft (M_0, A_0)$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$T_2 \quad \frac{\mathcal{E}(z : \sigma) \triangleleft (z, \{z : \sigma\})}{\mathcal{E}(I : \sigma \rightarrow \sigma) \triangleleft I}$$

$$T_3 \quad \frac{\mathcal{E}(z : \sigma) \triangleleft (z, \{z : \sigma\})}{\mathcal{E}(I : \sigma \rightarrow \sigma) \triangleleft I}$$

$$\frac{\frac{\frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\})}{\mathcal{E}(f(fx) : \sigma) \triangleleft ((f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})} T_4}{\mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (\lambda x.(f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\})} T_2 \quad T_3}{\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft \lambda fx.(f(fx^1)^1)} T_1$$

$$\frac{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft (M_1, A_1)}{\mathcal{E}((\lambda fx.f(fx))I : \sigma) \triangleleft (M_0, A_0)}$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$T_2 \quad \frac{\mathcal{E}(z : \sigma) \triangleleft (z, \{z : \sigma\})}{\mathcal{E}(I : \sigma \rightarrow \sigma) \triangleleft I}$$

$$T_3 \quad \frac{\mathcal{E}(z : \sigma) \triangleleft (z, \{z : \sigma\})}{\mathcal{E}(I : \sigma \rightarrow \sigma) \triangleleft I}$$

$$\begin{array}{c} \frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\})}{\mathcal{E}(f(fx) : \sigma) \triangleleft ((f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})} \quad T_4 \\ \frac{\mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (\lambda x.(f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\})}{\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft \lambda fx.(f(fx^1)^1)} \quad T_2 \quad T_3 \\ \frac{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft ((\lambda fx.(f(fx^1)^1))(I \mid I))}{\mathcal{E}((\lambda fx.f(fx))I : \sigma) \triangleleft (M_0, A_0)} \quad T_1 \end{array}$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

$$\begin{array}{c}
 \frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\}) \quad T_4}{\mathcal{E}(f(fx) : \sigma) \triangleleft ((f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})} \\
 \frac{\mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (\lambda x.(f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\})}{\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft \lambda fx.(f(fx^1)^1) \quad T_2 \quad T_3} \\
 \frac{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft ((\lambda fx.(f(fx^1)^1))I^2) \quad T_1}{\mathcal{E}((\lambda fx.f(fx))I : \sigma) \triangleleft (M_0, A_0)}
 \end{array}$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))ll$, where $l \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

T_1

$$\overline{\mathcal{E}(l : \sigma) \triangleleft (M_8, A_8)}$$

$$\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\}) \quad T_4$$

$$\overline{\mathcal{E}(f(fx) : \sigma) \triangleleft ((f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})}$$

$$\overline{\mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (\lambda x.(f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\})}$$

$$\overline{\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft \lambda fx.(f(fx^1)^1)} \quad T_2 \quad T_3$$

$$\overline{\mathcal{E}((\lambda fx.f(fx))l : \sigma \rightarrow \sigma) \triangleleft ((\lambda fx.(f(fx^1)^1))l^2)} \quad T_1$$

$$\mathcal{E}((\lambda fx.f(fx))ll : \sigma) \triangleleft (M_0, A_0)$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

T_1

$$\frac{\mathcal{E}(z : \alpha) \triangleleft (z, \{z : \alpha\})}{\mathcal{E}(I : \sigma) \triangleleft (M_8, A_8)}$$

$$\frac{\frac{\frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\}) \quad T_4}{\mathcal{E}(f(fx) : \sigma) \triangleleft ((f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})}}{\mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (\lambda x.(f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\})}}{\frac{\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft \lambda fx.(f(fx^1)^1) \quad T_2 \quad T_3}{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft ((\lambda fx.(f(fx^1)^1))I^2)} \quad T_1$$

$$\frac{}{\mathcal{E}((\lambda fx.f(fx))II : \sigma) \triangleleft (M_0, A_0)}$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

T_1

$$\frac{\mathcal{E}(z : \alpha) \triangleleft (z, \{z : \alpha\})}{\mathcal{E}(I : \sigma) \triangleleft I}$$

$$\frac{\frac{\frac{\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\})}{\mathcal{E}(f(fx) : \sigma) \triangleleft ((f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})} \quad T_4}{\mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (\lambda x.(f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\})} \quad T_2 \quad T_3}{\frac{\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft \lambda fx.(f(fx^1)^1)}{\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft ((\lambda fx.(f(fx^1)^1))I^2)} \quad T_1}{\mathcal{E}((\lambda fx.f(fx))II : \sigma) \triangleleft (M_0, A_0)}$$

Term Expansion with AC-intersection types

Consider the the term $(\lambda fx.f(fx))I$, where $I \equiv \lambda z.z$ and $\sigma \equiv \alpha \rightarrow \alpha$

T_1

$$\frac{\mathcal{E}(z : \alpha) \triangleleft (z, \{z : \alpha\})}{\mathcal{E}(I : \sigma) \triangleleft I}$$

$$\mathcal{E}(f : \sigma \rightarrow \sigma) \triangleleft (f, \{f : \sigma \rightarrow \sigma\}) \quad T_4$$

$$\mathcal{E}(f(fx) : \sigma) \triangleleft ((f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma), x : \sigma\})$$

$$\mathcal{E}(\lambda x.f(fx) : \sigma \rightarrow \sigma) \triangleleft (\lambda x.(f(fx^1)^1), \{f : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma)\})$$

$$\mathcal{E}(\lambda fx.f(fx) : (\sigma \rightarrow \sigma) \cap (\sigma \rightarrow \sigma) \rightarrow (\sigma \rightarrow \sigma)) \triangleleft \lambda fx.(f(fx^1)^1) \quad T_2 \quad T_3$$

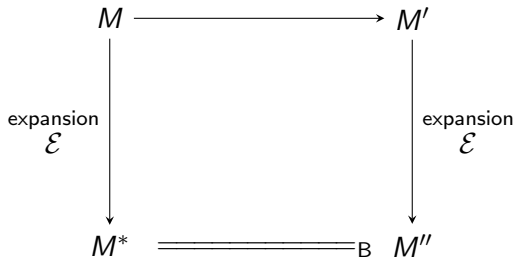
$$\mathcal{E}((\lambda fx.f(fx))I : \sigma \rightarrow \sigma) \triangleleft ((\lambda fx.(f(fx^1)^1))I^2) \quad T_1$$

$$\mathcal{E}((\lambda fx.f(fx))II : \sigma) \triangleleft (((\lambda fx.(f(fx^1)^1))I^2)I^1)$$

Expansion and Finite Multiplicities

Theorem

Given a λx -term M and an AC-intersection type σ , such that $\mathcal{E}(M : \sigma) \triangleleft (M^*, A_1)$, if $M \xrightarrow[bxgc]{\quad} M'$ then $M^* =_B M''$ and $\mathcal{E}(M' : \sigma) \triangleleft (M'', A_2)$, where $A_2 \subseteq A_1$.



Illustrative Example

$$(\lambda fx.f(fx))//$$

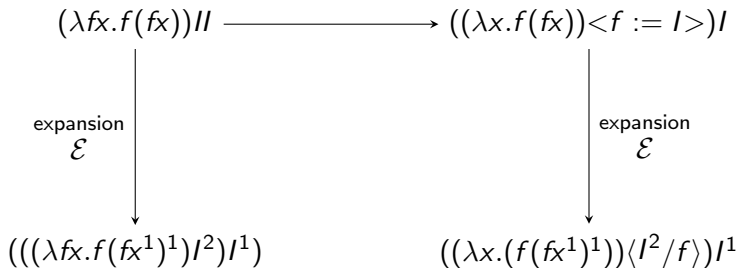
Illustrative Example

$$(\lambda fx.f(fx))I \longrightarrow ((\lambda x.f(fx))\langle f := I \rangle)I$$

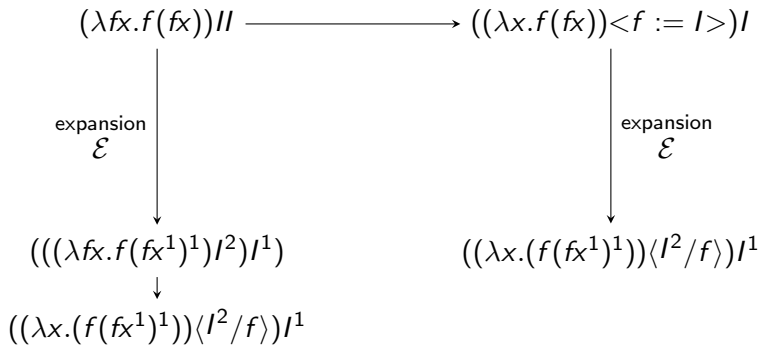
Illustrative Example

$$\begin{array}{ccc} (\lambda f x. f(fx))I & \longrightarrow & ((\lambda x. f(fx))\langle f := I \rangle)I \\ \downarrow \text{expansion } \mathcal{E} & & \\ (((\lambda f x. f(fx^1)^1)I^2)I^1) & & \end{array}$$

Illustrative Example



Illustrative Example



Illustrative Example

$$\begin{array}{ccc}
 (\lambda f x. f(fx))I & \longrightarrow & ((\lambda x. f(fx))\langle f := I \rangle)I \\
 \downarrow \text{expansion } \mathcal{E} & & \downarrow \text{expansion } \mathcal{E} \\
 (((\lambda f x. f(fx^1)^1)I^2)I^1) & \equiv_B & ((\lambda x. (f(fx^1)^1))\langle I^2/f \rangle)I^1 \\
 \downarrow & & \\
 ((\lambda x. (f(fx^1)^1))\langle I^2/f \rangle)I^1 & &
 \end{array}$$

Final Remarks

- ▶ We proved that
 - ▶ ACI-intersection types $\Rightarrow$ a resource calculus that deals with infinite multiplicities
 - ▶ AC-intersection types $\Rightarrow$ a resource calculus that deals with finite multiplicities
- ▶ In the future, we want to study the application of *term expansion* to relate the λ -calculus with concurrent calculi

Final Remarks

- ▶ We proved that
 - ▶ ACI-intersection types $\Rightarrow$ a resource calculus that deals with infinite multiplicities
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- ▶ In the future, we want to study the application of *term expansion* to relate the λ -calculus with concurrent calculi

Thank you